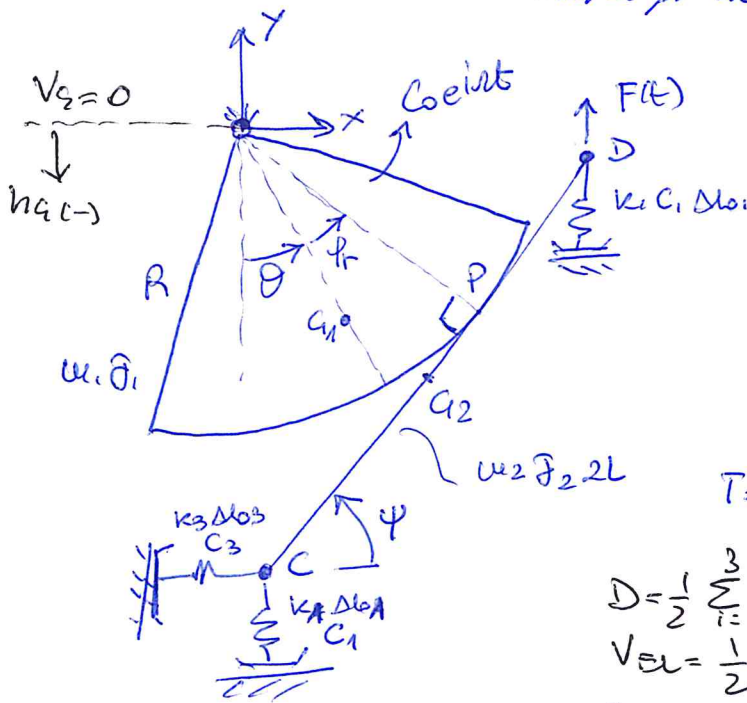


Standard problem



$$\underline{x} = \begin{Bmatrix} \theta \\ \phi \end{Bmatrix} \quad \underline{x}_0 = \begin{Bmatrix} \theta_0 \\ \phi_0 \end{Bmatrix}$$

$$T = \frac{1}{2} \omega_1 \dot{\theta}_1^2 + \frac{1}{2} \dot{\theta}_1^2 + \frac{1}{2} \omega_2 \dot{\phi}_2^2 + \frac{1}{2} \dot{\phi}_2^2 + \frac{1}{2} \dot{\phi}_2^2$$

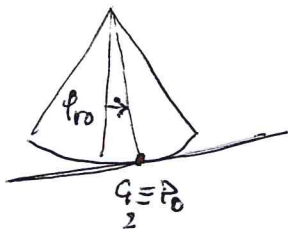
$$D = \frac{1}{2} \sum_{i=1}^3 C_i \dot{\Delta l}_i^2$$

$$V_{el} = \frac{1}{2} \sum_{i=1}^3 k_i \Delta l_i^2 \quad V_g = \omega_1 g h_{g1} + \omega_2 g h_{g2}$$

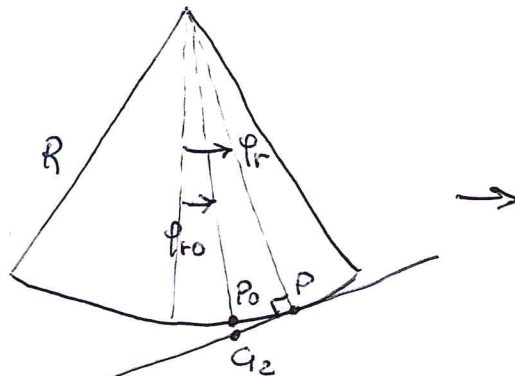
$$\delta W = \text{Coefint} \delta \theta + F(t) \delta y_D$$

Rolling without sliding condition:

A) static configuration



B) dynamic conf.



→ the arc $\widehat{PP_0} = R(\phi_r - \phi_{r0})$ is equal to the distance $\widehat{G_2P}$ →
 $\widehat{G_2P} = R(\phi_r - \phi_{r0})$

$$x_p = R \sin(\theta + \phi_r)$$

$$y_p = -R \cos(\theta + \phi_r)$$

$$\bar{PD} = L - R(\phi_r - \phi_{r0})$$

$$\bar{PC} = L + R(\phi_r - \phi_{r0})$$

$$\psi = \theta + \phi_r$$

$$y_D = y_p + \bar{PD} \sin \psi$$

$$y_c = y_p - \bar{PC} \sin \psi$$

$$x_c = x_p - \bar{PC} \cos \psi$$

$$\begin{cases} h_{g1} = y_{g1} = -r \cos \theta \\ h_{g2} = y_{g2} \end{cases}$$

$$y_D = -R \cos(\theta + \phi_r) + L \sin(\theta + \phi_r) - R(\phi_r - \phi_{r0}) \sin(\theta + \phi_r)$$

$$y_c = -R \cos(\theta + \phi_r) - L \sin(\theta + \phi_r) - R(\phi_r - \phi_{r0}) \sin(\theta + \phi_r)$$

$$x_c = R \sin(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) - L \cos(\theta + \varphi_r)$$

$$x_{q2} = R \sin(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r)$$

$$y_{q2} = -R \cos(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r)$$

$$\Delta L_1 = y_c - y_{c0} ; \Delta L_2 = y_D - y_{D0} ; \Delta L_3 = x_c - x_{c0}$$

$$\begin{cases} \frac{\partial x_{q2}}{\partial \theta} = R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \\ \frac{\partial x_{q2}}{\partial \varphi_r} = R \cos(\theta + \varphi_r) - R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \end{cases}$$

$$\begin{cases} \frac{\partial y_{q2}}{\partial \theta} = R \sin(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) \\ \frac{\partial y_{q2}}{\partial \varphi_r} = R \sin(\theta + \varphi_r) - R \sin(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) \end{cases}$$

$$\begin{cases} \frac{\partial \Delta L_1}{\partial \theta} = R \sin(\theta + \varphi_r) - L \cos(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) \\ \frac{\partial \Delta L_1}{\partial \varphi_r} = R \sin(\theta + \varphi_r) - L \cos(\theta + \varphi_r) - R \sin(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) \end{cases}$$

$$\begin{cases} \frac{\partial \Delta L_2}{\partial \theta} = R \sin(\theta + \varphi_r) + L \cos(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) \\ \frac{\partial \Delta L_2}{\partial \varphi_r} = R \sin(\theta + \varphi_r) + L \cos(\theta + \varphi_r) - R \sin(\theta + \varphi_r) - R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) \end{cases}$$

$$\frac{\partial \Delta L_3}{\partial \theta} = R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) + L \sin(\theta + \varphi_r)$$

$$\frac{\partial \Delta L_3}{\partial \varphi_r} = R \cos(\theta + \varphi_r) - R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) + L \sin(\theta + \varphi_r)$$

$$\begin{aligned} [A_{gen}] &= \begin{bmatrix} r & 0 \\ 1 & 0 \\ R \cos \varphi_{r0} & 0 \\ R \sin \varphi_{r0} & 0 \\ 1 & 1 \end{bmatrix} \begin{matrix} v_{z1} \\ \theta \\ x_{q2} \\ y_{q2} \\ \varphi \end{matrix} & [A_d] = [A_{gen}] &= \begin{bmatrix} R \sin \varphi_{r0} - L \cos \varphi_{r0} & -L \cos \varphi_{r0} \\ R \sin \varphi_{r0} + L \cos \varphi_{r0} & L \cos \varphi_{r0} \\ R \cos \varphi_{r0} + L \sin \varphi_{r0} & L \sin \varphi_{r0} \end{bmatrix} \\ [A_e] &= \begin{bmatrix} 1 & 0 \\ R \sin \varphi_{r0} + L \cos \varphi_{r0} & L \cos \varphi_{r0} \end{bmatrix} \end{aligned}$$

$$\begin{cases} \frac{\partial^2 \psi_{s2}}{\partial \theta^2} = R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \\ \frac{\partial^2 \psi_{s2}}{\partial \theta \partial \varphi_r} = R \cos(\theta + \varphi_r) - R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \\ \frac{\partial^2 \psi_{s2}}{\partial \varphi_r^2} = R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) - R \cos(\theta + \varphi_r) \end{cases}$$

$$\frac{\partial^2 \psi_{s1}}{\partial \theta^2} = R \cos \theta \quad [k_{s1}] = m_2 g \begin{bmatrix} r & 0 \\ 0 & 0 \end{bmatrix}$$

$$[k_{s2}] = m_2 g \begin{bmatrix} R \cos \varphi_{r0} & 0 \\ 0 & -R \cos \varphi_{r0} \end{bmatrix}$$

$$\begin{cases} \frac{\partial^2 \Delta \psi_1}{\partial \theta^2} = R \cos(\theta + \varphi_r) + L \sin(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \\ \frac{\partial^2 \Delta \psi_1}{\partial \theta \partial \varphi_r} = R \cos(\theta + \varphi_r) + L \sin(\theta + \varphi_r) - R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \\ \frac{\partial^2 \Delta \psi_1}{\partial \varphi_r^2} = L \sin(\theta + \varphi_r) - R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \end{cases}$$

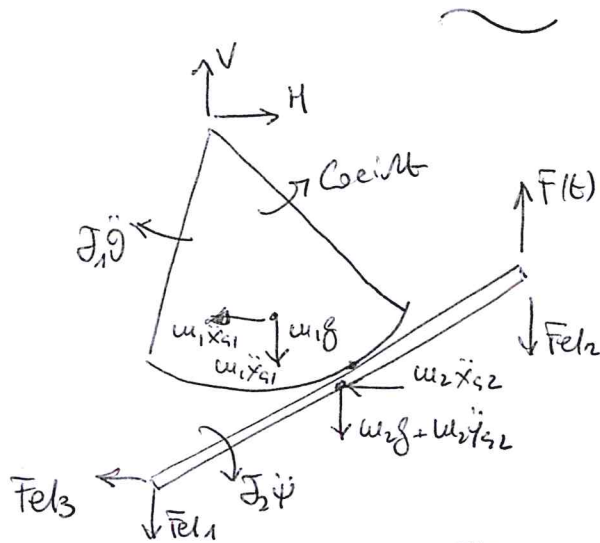
$$\begin{cases} \frac{\partial^2 \Delta \psi_2}{\partial \theta^2} = R \cos(\theta + \varphi_r) - L \sin(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \\ \frac{\partial^2 \Delta \psi_2}{\partial \theta \partial \varphi_r} = R \cos(\theta + \varphi_r) - L \sin(\theta + \varphi_r) - R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \\ \frac{\partial^2 \Delta \psi_2}{\partial \varphi_r^2} = -L \sin(\theta + \varphi_r) - R \cos(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \sin(\theta + \varphi_r) \end{cases}$$

$$\begin{cases} \frac{\partial^2 \Delta \psi_3}{\partial \theta^2} = -R \sin(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) + L \cos(\theta + \varphi_r) \\ \frac{\partial^2 \Delta \psi_3}{\partial \theta \partial \varphi_r} = -R \sin(\theta + \varphi_r) + R \sin(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) + L \cos(\theta + \varphi_r) \\ \frac{\partial^2 \Delta \psi_3}{\partial \varphi_r^2} = R \sin(\theta + \varphi_r) + R(\varphi_r - \varphi_{r0}) \cos(\theta + \varphi_r) + L \cos(\theta + \varphi_r) \end{cases}$$

$$[K_{EL, II 1}] = k_1 \Delta \phi_1 \begin{bmatrix} R \cos \varphi_0 + L \sin \varphi_0 & L \sin \varphi_0 \\ L \sin \varphi_0 & L \sin \varphi_0 - R \cos \varphi_0 \end{bmatrix}$$

$$[K_{EL, II 2}] = k_2 \Delta \phi_2 \begin{bmatrix} R \cos \varphi_0 - L \sin \varphi_0 & -L \sin \varphi_0 \\ -L \sin \varphi_0 & -L \sin \varphi_0 - R \cos \varphi_0 \end{bmatrix}$$

$$[K_{EL, II 3}] = k_3 \Delta \phi_3 \begin{bmatrix} -R \sin \varphi_0 + L \cos \varphi_0 & L \cos \varphi_0 \\ L \cos \varphi_0 & R \sin \varphi_0 + L \cos \varphi_0 \end{bmatrix}$$



$$\sum F_x = 0: H - m_1 \ddot{x}_{s1} - m_2 \ddot{x}_{s2} - F_{el3} = 0$$

$$H = m_1 \ddot{x}_{s1} + m_2 \ddot{x}_{s2} + F_{el3}$$